Module-1 : Understanding of hardware and its components

Section 1: Multiple Choice

Q1. Which of the following is NOT a component of CPU?

1. ALU
2. RAM
3. CU
4. 1 and 3 both

Answer : RAM

Q2. What Is the function of RAM in a computer?

Answer : RAM (Random Access Memory) acts as the computer short-term memory. It provides high speed storage for data and instruction that the CPU is currently using.

Q3. Which of the following is a primary storage device?

1. HDD
2. SSD
3. SD Card
4. 1-2 both

Answer : 1-2 both

Q4. What is the purpose of a GPU?

Answer : GPU (Graphical Processing Unit) is a specialized processor. GPU is designed to handle many mathematical tasks simultaneously.

Section 2: True or False

Q1. The motherboard is the main circuit board of a computer where other components are attached.

Answer : True

Q2. A UPS (uninterruptible Power Supply) is a hardware device that provides emergency power to a load when the input power source fails.

Answer : True

Q3. An expansion card is a circuit board that enhances the functionality of a component.

Answer : True

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Section 3 : Short Answer

Q1. Explain the difference between HDD and SSD.

Answer :

Features	**HDD**	**SSD**
Speed	Slower than SSD (30-150MB/s)	Faster than HDD (500-700 MB/s)
Durability	Less durable	More durable
Weight	Heavy in weight	Light in weight
Power	Consume more power	Consume less power
Cost	Cheaper than SSD	More expensive

Q2. Describe the function of BIOS in a computer system.

Answer : The BIOS (Basic Input/Output) is a firmware used to perform hardware initialization during the booting process.

Q3. List and briefly explain three input devices commonly used with computers.

Answer :

1. Keyboard : A keyboard is used to enter text, numbers, and commands into a computer.
   It allows users to type documents, emails, and instructions for software.
   Special keys help perform shortcuts and control system functions.
2. Mouse : A mouse is a pointing device used to move the cursor on the screen.
   It helps users select, click, drag, and interact with graphical elements.
   This makes navigation faster and more intuitive.
3. Microphone : It allows users to record voice, music, or other audio.
   Microphones are commonly used for online meetings, voice commands, and gaming.

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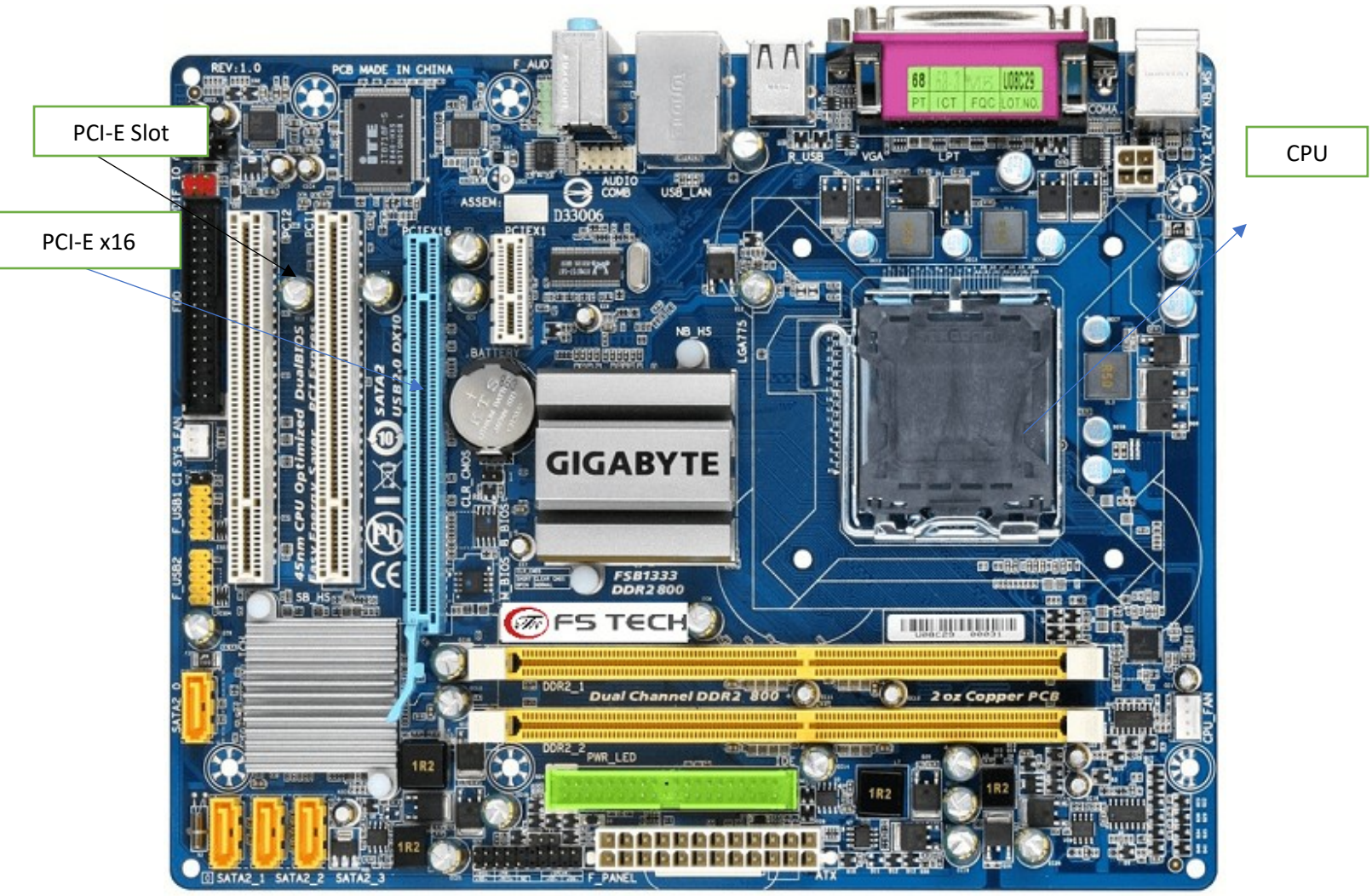
Section 4 : Practical Application

Q1. Identify and label the following components on a diagram of a motherboard.

- CPU
- RAM
- SATA Connector
- PCI-E Slot

Answer :



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RAM :

SATA Connector :

Q2. Demonstrate how to install a RAM module into a computer.

Answer :

1. Power off and unplug : Shut down your computer completely.
2. Open the case : remove the side panel and for laptop, remove the bottom panel.
3. Open the RAM slot : push the small clip on each end of the slot outward.
4. Align the RAM : align the RAM module match it with the slot.
5. Insert the RAM : press the RAM lightly down until the clip snap.
6. Verify the installation : go to Task manager->Performance->Memory, here you can see your installed RAM.

Section 5 : Essav

Q1. Discuss the importance of proper cooling mechanisms in a computer system. Include examples of cooling methods and their effectiveness.

Answer :

Importance of proper cooling

1. **Prevent overheating :** Components like the CPU, GPU, and power supply produce a lot of heat. Proper cooling keeps their temperature within safe limits.
2. **Improve performance :** When a system overheats, it may slow down automatically (thermal throttling). Good cooling helps the computer run at full speed.
3. **Increases hardware lifespan :** Continuous high temperature can permanently damage components. Cooling reduces wear and tear, increasing durability.
4. **Ensure system stability :** Overheating can cause sudden shutdowns, crashes, or data loss. Cooling helps maintain stable operation.
5. **Energy efficiency :** Efficient cooling allows components to work optimally, reducing unnecessary power consumption.

Examples of Cooling Methods and Their Effectiveness

1. **Air Cooling (Fans & Heat Sinks)**
   - Uses fans to circulate air and heat sinks to absorb heat.
   - Effectiveness: Cost-effective and suitable for most normal computers.
2. **Liquid Cooling (Water Cooling)**
   - Uses liquid to transfer heat away from components.
   - Effectiveness: Very effective for high-performance systems and gaming PCs.
3. **Thermal paste**
   - Applied between the CPU and heat sink to improve heat transfer
   - Effectiveness: Enhances cooling efficiency significantly.
4. **Passive cooling**
   - Uses heat sinks without fans
   - Effectiveness: Silent but less effective; suitable for low-power devices.
5. **Ventilation and Case Design**

- Proper airflow through cabinet design.
- Effectiveness: Supports all cooling methods and prevents heat buildup.

Q2. Explain the concept of bus width and its significance in computer architecture.

Answer :

Bus width concept

Bus width refers to the number of bits that a data bus can carry at one time between computer. components such as the CPU, memory, and I/O devices. For example, a 32-bit bus can transfer 32 bits of data simultaneously, while a 64-bit bus can transfer 64 bits at once.

Significance of Bus Width in Computer Architecture

1. **Data transfer speed**
   - A wider bus can carry more data in a single cycle, which increases overall system performance.
2. **Processor performance**
   - CPUs with larger bus widths can process larger chunks of data faster, making them more powerful.
3. **Memory access**
   - Greater bus width allows the CPU to fetch more data from memory at one time, reducing delays
4. **System capability**
   - Bus width affects how much data the system can handle efficiently, especially in multitasking and heavy applications.